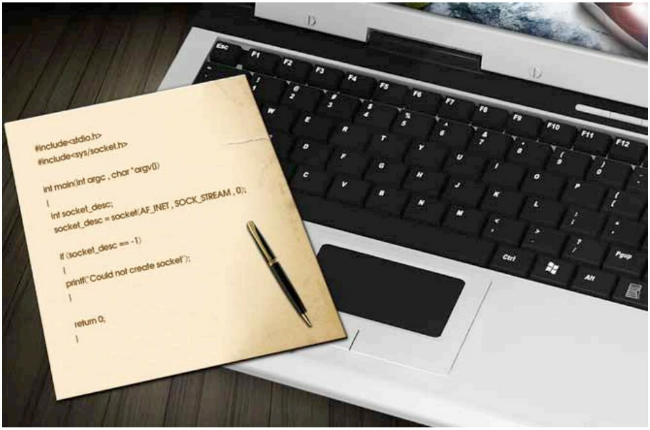
Programming Socket in C (TCP)

This article explores the basics of writing programs in C using BSD 'sockets' in the IP version 4 protocol. The article published in the January 2013 issue of *OSFY* covered UDP socket programming; here, we take a look at socket programs using TCP (Transmission Control Protocol).



Consider a network application—say, a file transfer program. It has a client and server. The client connects to the server to request files for download, or to transfer files to the server (upload). Such applications need to conform to some protocol; the most common is TCP/IP, which is segregated into layers like the application, transport, network and link layers. Sockets help us get the services of the transport layer (TCP, UDP or SCTP sockets). Using RAW sockets, you can directly get the services of the network layer.

With TCP, first a connection is established, then data is transferred, and then the connection is terminated. To establish a connection, three packets are exchanged between client and server, and after this 'handshake', data transfer takes place. In normal scenarios, one side initiates connection termination; and four packets are exchanged for a proper termination.

What does a socket program look like?

A typical socket program would have two source files, one for the server and one for the client. Our files are named *lfy_tcp_client.c* and *lfy_tcp_server.c*. In this example, there is another file, *functions.c*, which contains some common utility functions for both client and server. Please download these from Github, *https://github.com/mybodhizone/lfy_tcp_socket*, after which they need to be compiled to make executables—say, *myclient* and *myserver*. Run these in two terminals, as shown below:

```
[localhost tcp]$ gcc lfy_tcp_server.c functions.c -o myserver
[localhost tcp]$ ./myserver
TCP Server: Waiting for new client connection
Server: Connection from client -127.0.0.1
Server: Received from client:hello
```

```
[localhost tcp]$ gcc lfy_tcp_client.c functions.c -o myclient
[localhost tcp]$ ./myclient 127.0.0.1 11710
Client - Trying to connect to server
Client - Connected Successfully to server
Enter Data For the server or press CTRL-D to exit
hello
```